

MANINGRIDA, Australia

WMO: 951420

Lat: 12.0569S

Lon: 134.2339E

Elev: 97

StdP: 14.64

Time Zone: 9.50 (AUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	61.3	63.7	44.4	43.3	76.8	47.8	49.3	76.1	19.3	79.3	18.4	80.3	5.5	170	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.1	92.9	77.2	91.9	77.1	91.0	76.9	82.2	88.2	81.0	87.3	80.3	86.7	10.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.7	160.2	84.9	79.5	153.8	84.2	78.7	149.5	83.7	46.2	88.5	44.9	87.5	44.1	86.8	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	16.1	14.5	DB	55.7	96.1	3.0	1.0	53.5	96.9	51.8	97.5	50.1	98.1	48.0	98.8	(4)
(5)			WB	51.2	84.4	2.7	3.4	49.3	86.9	47.7	88.9	46.2	90.8	44.3	93.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.5	83.1	83.0	82.3	81.6	79.9	76.7	75.8	75.8	78.5	81.4	83.8	84.2
	DBStd	3.78	2.16	2.10	1.95	1.79	2.47	3.45	2.93	2.63	2.35	2.05	1.50	2.01
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11132	1026	923	1002	948	927	800	801	800	855	974	1014	1062
	CDD65	5658	561	503	537	498	462	351	336	335	405	509	564	597
	CDH74	61006	6430	5674	5743	5236	4490	2880	2733	2979	4070	5926	7159	7686
	CDH80	23377	2470	2139	2048	1958	1569	854	799	929	1412	2407	3261	3531
Wind	WSAvg	7.4	6.8	6.7	6.4	7.0	8.1	9.0	9.0	7.9	7.2	7.0	7.2	6.9
Precipitation	PrecAvg	50.6	11.1	10.9	11.5	5.0	0.9	0.0	0.0	0.0	0.0	0.5	2.2	8.8
	PrecMax	85.6	27.8	21.7	48.2	28.9	8.0	0.4	0.4	0.1	0.4	2.8	11.4	23.7
	PrecMin	27.1	2.5	1.7	2.8	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5
	PrecStd	14.8	5.2	4.7	8.3	5.0	1.6	0.1	0.1	0.0	0.1	0.7	2.6	6.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	93.0	93.3	92.3	91.2	89.2	89.2	90.0	91.9	93.2	93.3	94.4
		MCWB	79.8	78.9	78.2	74.9	74.2	71.8	69.1	67.2	68.1	71.8	77.0	79.0
	2%	DB	91.7	91.2	90.8	90.9	89.6	87.7	87.4	88.1	89.6	90.9	91.8	92.7
		MCWB	79.8	79.3	78.1	75.1	74.0	71.5	69.2	67.7	71.0	74.6	77.0	78.4
	5%	DB	90.1	89.9	89.2	89.7	88.4	86.3	86.1	86.5	88.0	89.7	90.9	91.6
		MCWB	79.4	79.1	78.3	75.4	73.5	70.9	69.2	67.9	71.5	75.1	76.9	78.4
	10%	DB	88.6	88.6	88.1	88.4	87.2	85.0	84.6	85.1	86.7	88.6	90.0	90.5
		MCWB	79.2	78.8	78.3	75.6	73.2	70.3	68.8	68.1	71.7	74.8	76.8	78.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	85.2	82.4	83.3	80.3	79.2	75.9	74.7	74.6	77.0	79.1	80.8	83.0
		MCDB	88.4	88.8	87.5	86.2	85.4	83.6	82.7	82.8	84.2	87.4	89.1	89.6
	2%	WB	82.3	81.0	81.3	79.3	78.1	74.9	73.6	73.6	75.9	77.8	79.7	81.2
		MCDB	88.3	87.5	86.5	85.6	84.4	82.7	81.2	81.5	83.5	86.5	88.2	88.7
	5%	WB	81.0	80.2	80.3	78.6	77.1	73.8	72.7	72.8	75.1	77.0	78.9	80.2
		MCDB	86.7	86.7	85.7	84.9	83.3	81.9	80.5	80.7	83.1	85.8	87.7	87.9
	10%	WB	80.0	79.6	79.5	77.9	76.1	72.8	71.7	71.8	74.3	76.4	78.1	79.4
		MCDB	85.9	86.1	85.0	84.2	82.8	81.0	80.0	79.8	82.4	85.1	87.0	87.3
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.1	11.6	13.9	15.8	17.5	19.1	20.5	18.3	15.7	13.7	12.1
		MCDBR	13.3	13.0	13.5	15.9	16.6	17.5	19.6	21.9	19.6	16.4	14.3	13.7
		MCWBR	5.6	5.4	5.0	4.8	5.4	6.1	7.1	8.8	7.4	5.5	4.4	4.5
	5% WB	MCDBR	11.8	11.9	11.9	12.8	14.0	15.6	16.9	17.6	15.7	14.7	13.4	12.5
	MCWBR	6.4	5.6	5.4	4.6	5.4	5.8	7.0	7.8	5.8	5.2	5.2	5.6	
Clear-Sky Solar Irradiance	taub	0.411	0.402	0.391	0.379	0.372	0.357	0.350	0.360	0.407	0.426	0.426	0.408	
	taud	2.531	2.553	2.586	2.599	2.591	2.595	2.604	2.552	2.409	2.380	2.424	2.522	
	Ebn at Noon	296	296	294	288	280	279	284	290	285	286	289	296	
	Edh at Noon	36	35	33	30	29	28	29	31	38	41	39	36	
All-Sky Solar Radiation	RadAvg	1686	1738	1703	1737	1656	1585	1673	1912	2120	2212	2161	1880	
	RadStd	203	250	192	203	114	74	61	62	80	92	90	178	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (7 neighbors)	+0.72	N/A	N/A	N/A	N/A	-0.85	N/A	N/A	N/A	N/A

Nomenclature: See separate page